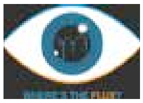
First-ever 4K video of Mars
YouTubers, ElderFox Documentaries, used several pictures of Mars from 3 rovers to create a 4K video of the Red Planet. **https://dgit.in/aug20-03**

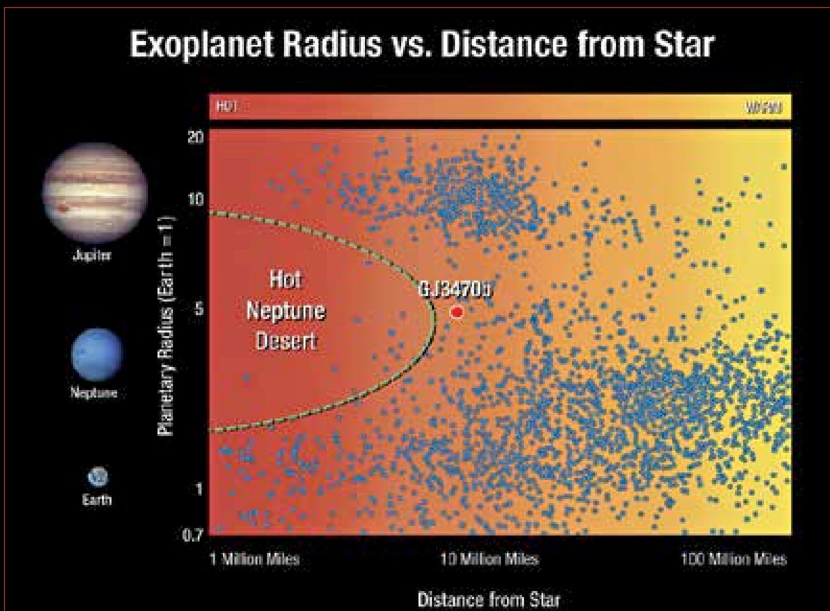


A place like home
Scientists captured the first image of a solar system that resembles our own, about 300 light years away. **https://dgit.in/aug20-04**

planets found between these two extremes, planets known as Hot Neptunes. Note that the Neptune Desert exists only for planets with short orbital periods, Neptune-sized Ice Giants with longer orbits are fairly abundant. There was no reason why mid-sized planets should not exist in close orbits, and the Hot Neptune desert was something of a mystery.

There have been a few Hot Neptunes discovered, however, the first of these was found in 2004. The planet was discovered by precisely measuring the way its gravity made the host star wobble. This planet, known as Gliese 436 b was the first of its kind to be found. Since then a few planets on the edge of the Hot Neptune desert have been found including TOI-132 b, LTT 9779b, GJ 3470 b and GJ 436 b. In all of these worlds, astronomers found that the energetic action of the stars was rapidly stripping away the atmospheres of the planets, and they were all losing mass at a rapid rate. If the stars were younger, the loss in mass was even more pronounced, considering that young stars are more energetic. Here was an explanation for the lack of Neptune-sized worlds in close orbit around host stars, unless they were discovered during the period when they were being stripped of most of their mass, they would just be turned into planets that would resemble rocky planets, and from a distance, would be similar in characteristics to a class of planets known as Super Earths.

AU Microscopii is a relatively nearby star system, about 32 light years away. The host planet is a red dwarf, which is only 12 million years old, or one percent the age of our Sun. This planet has a protoplanetary disc surrounding it, and has been a target of observation campaigns as it promises to provide incredible insight into the early periods of our own Solar System, as well as other star systems. The process of planetary formation is very much ongoing here. In July this year, scientists reported that they have found a Neptune-sized body in this system, the first identified planet around the star, called AU Mic in short. The planet, named AU Mic b has a year that lasts only 8.5 days. This discovery has been particularly challenging as very young stars have periodic flares, as well as numerous sunspots, both of which change the brightness of profiles rapidly, disrupting the observations needed to identify exoplanets in orbit around them.



Credit: NASA

The Hot Neptune desert

Close on the heels of that discovery, astronomers found another planet that is even more interesting. In orbit around a star 750 light years away, is the planet TOI 849 b, which is about 3.5 times larger than the Earth, about the size of Neptune. It definitely qualifies as a Hot Neptune, as the orbit is so close to the host star, that an entire year lasts only 18 hours! The planet is about 40 times as massive as the Earth, and just as dense. Most of the Neptune-sized planets found so far, have a lesser density than rocky planets. TOI 849 b is the most massive planet of its kind discovered so far, as well as the one with the highest density. These unique properties makes scientists believe that TOI 849 b is very special, in the sense it is what remains of a Neptune-like ice giant, after all the material has been stripped away by the host star. The star was found by an astronomer serendipitously who looked for the telltale signatures of light dips by transiting exoplanets, in the data captured from the TESS telescopes, for stars farther away than those normally studied by scientists. The discovery was followed up by ground-based observations that confirmed the existence of the planet. Current planetary models cannot explain how the planet got so massive, and remained that way without getting even more massive. However, if the planet was even more massive, and then was stripped of the mass, then the observations make sense. If confirmed, this is the first discovery of a naked planetary core. It is a large body, and further studies on the object can inform us about the inner structure of the gas giants and ice giants in our own Solar System, something that scientists know precious little about. d